CONJURA · COMPANION NOTE

What the Leftover Hash Lemma Gives

And What It Cannot: a Limitation on the Collision Route

Resolves. Nothing. This is a companion note to *Randomness Extraction from Unpredictable Random-Oracle Sources: A Leftover-Hash-Lemma Conjecture, Public Seed* (`main.tex` in this folder), whose conjecture is proved in `proof.tex` beside it. Two things are recorded. On a fixed universal table the leftover hash lemma returns the conjectured bound's first summand exactly (Proposition 2). No table available at the seed lengths at issue is universal, and on the tables that are available the lemma returns *at least* $\frac{1}{2}\sqrt{(W-1)/K}$ for a source it cannot exclude, where W is defined in Theorem 1 and is of order $\min\{R, D\epsilon\}$; under the side condition of Lemma 1 that floor is a statement about the conjecture's *second* summand, where it costs $\log_2 W$ bits of seed against the conjectured $\log_2 \log_2 D$. Section 8 says what is *not* claimed, and in particular that no matching upper bound is proved: the floor is one-sided.

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1 Setting

The setting is that of `main.tex`, repeated here so this note stands alone. Fix nonempty finite sets $\mathcal{K}$ (seeds), $\mathcal{D}$ (inputs), $\mathcal{R}$ (outputs), and write $K := |\mathcal{K}|$, $D := |\mathcal{D}|$, $R := |\mathcal{R}|$. A table H is drawn uniformly from $\mathsf{Fun}(\mathcal{K}\times\mathcal{D},\mathcal{R})$. A source S, given the *entire* table, outputs a pair $(x,z)\in\mathcal{D}\times\{0,1\}^*$; a seed $sd \xleftarrow{\$} \mathcal{K}$ is drawn *after* S has terminated; and an unbounded distinguisher D, given (H, sd, y_b, z), is to tell $y_0 = H(sd,x)$ from a uniform y_1. Write $\mathbf{Adv}^{\text{ext-pub}}(S,\mathsf{D})$ for twice its winning probability minus one. The source is *ϵ-unpredictable* if no unbounded predictor given (H,z) guesses x with probability more than ϵ; equivalently, if $\mathbb{E}_{(H,Z)}[\epsilon_{H,Z}] \le \epsilon$ in the notation below. Conjecture 1 of `main.tex` asserts a universal constant c such that, for every ϵ-unpredictable S and every unbounded D,

$$\mathbf{Adv}^{\text{ext-pub}}(S,\mathsf{D}) \;\le\; c\left(\sqrt{\epsilon R} + \sqrt{\frac{\log_2 D}{K}}\right). \tag{1}$$

Every numerical comparison below uses (1) exactly as displayed, with $c = 2$, the constant `proof.tex` establishes. No other constant from that note is imported and none is needed.

Throughout, condition on a table and an auxiliary output, $(H,Z) = (h,z)$, and write $p_x := \Pr[X = x \mid h, z]$ for the source's conditional law and $\epsilon_{h,z} := \max_x p_x$. The quantities

$$\mathrm{CP} \;:=\; \sum_{x\in\mathcal{D}} p_x^2, \qquad c_k \;:=\; \Pr_{X,X'}\big[h(k,X) = h(k,X')\big], \qquad \bar{c} \;:=\; \frac{1}{K}\sum_{k\in\mathcal{K}} c_k,$$

all depend on (h,z), which the notation suppresses; where they appear inside an expectation over (H,Z) they are to be read as the corresponding random variables. Here X,X' are independent draws from p. Note $\mathrm{CP} \le \epsilon_{h,z}$, since $\sum_x p_x^2 \le (\max_x p_x)\sum_x p_x$.

Fact 1 (Cauchy–Schwarz, two forms). Let q be a probability distribution on a finite set Ω and let $A \subseteq \Omega$ carry all of its mass. Then

$$\sum_\omega q_\omega^2 \;\ge\; \frac{1}{|A|}. \tag{2}$$

If moreover u is uniform on Ω, then

$$\mathrm{SD}(q,u) \;=\; \tfrac12\|q-u\|_1 \;\le\; \tfrac12\sqrt{|\Omega|}\,\|q-u\|_2 \;=\; \tfrac12\sqrt{|\Omega|\textstyle\sum_\omega q_\omega^2 - 1}\,. \tag{3}$$

Proof. For (2), $1 = (\sum_{\omega\in A} q_\omega)^2 \le |A|\sum_{\omega\in A} q_\omega^2 \le |A|\sum_\omega q_\omega^2$. For (3), the inequality is $\|v\|_1 \le \sqrt{|\Omega|}\,\|v\|_2$ applied to $v = q-u$, and the final equality is $\|q-u\|_2^2 = \sum_\omega q_\omega^2 - 1/|\Omega|$, which follows by expanding the square and using $\sum_\omega q_\omega = 1$. □

2 The leftover-hash step

Proposition 1 (The bound the lemma returns). *For every table h and every view z of positive probability,*

$$\max_{\mathsf{D}}\,\big(2\Pr[\mathsf{D}\ \textit{wins} \mid h,z] - 1\big) \;\le\; \tfrac12\sqrt{R\,\bar c - 1}\,,$$

and consequently, for every unbounded D, $\mathbf{Adv}^{\text{ext-pub}}(S,\mathsf{D}) \le \mathbb{E}_{(H,Z)}\big[\tfrac12\sqrt{R\,\bar c - 1}\big]$.

Proof. Because sd is drawn only after S has terminated, it is uniform on $\mathcal{K}$ and independent of X given (h,z). The distinguisher's view therefore differs between the two challenge bits only in the pair (sd, y_b), which is distributed as $Q := \mathrm{law}(sd, h(sd,X))$ under $b=0$ and as the uniform distribution U on $\mathcal{K}\times\mathcal{R}$ under $b=1$. The optimal advantage of an unbounded distinguisher between two distributions is their statistical distance, so the left-hand side is $\mathrm{SD}(Q,U)$.

Now $|\mathcal{K}\times\mathcal{R}| = KR$ and

$$\sum_{k,r} Q(k,r)^2 \;=\; \Pr\big[(sd,h(sd,X)) = (sd',h(sd',X'))\big] \;=\; \sum_k \frac{1}{K^2}\,c_k \;=\; \frac{\bar c}{K},$$

the middle equality because sd, sd' are independent and uniform, so both equal a given k with probability $1/K^2$, and given $sd = sd' = k$ the outputs collide with probability c_k. Feeding this into (3) gives $\mathrm{SD}(Q,U) \le \frac12\sqrt{KR\cdot\bar c/K - 1} = \frac12\sqrt{R\bar c - 1}$, the radicand being nonnegative by (2) applied to Q with $A = \mathcal{K}\times\mathcal{R}$. Averaging over (H,Z) gives the second claim. □

Writing $c_k = \mathrm{CP} + \beta_k$ with $\beta_k := \sum_{x\ne x'} p_x p_{x'}\mathbf{1}[h(k,x) = h(k,x')]$ and $\bar\beta := \frac1K\sum_k \beta_k$ splits the quantity under the root as

$$R\bar c - 1 \;=\; R\cdot\mathrm{CP} \;+\; \gamma, \qquad \gamma \;:=\; R\bar\beta - 1, \tag{4}$$

the first summand carrying the source's entropy deficiency and the second the family's failure to be universal on the source's support. Which of the two a lower bound on $R\bar{c} - 1$ speaks about is exactly the question Lemma 1 settles, and it must be settled: a bound on a sum is not by itself a bound on either part.

3 Calibration: on a universal table the lemma is sharp

Proposition 2 (Universal families). *Fix a table h and suppose the family $\{h(k,\cdot)\}_{k\in\mathcal{K}}$ is 2-universal, that is $\Pr_k[h(k,x) = h(k,x')] \le 1/R$ for all $x \ne x'$. Then $\gamma \le -\mathrm{CP} \le 0$ for every source law p, and the conditional bound of Proposition 1 is at most $\frac{1}{2}\sqrt{R\,\epsilon_{h,z}}$. Consequently, if* every *table in the support of H has this property, then $\mathbf{Adv}^{\text{ext-pub}}(S, \mathsf{D}) \le \frac{1}{2}\sqrt{\epsilon R}$ for every ϵ-unpredictable S and every unbounded D.*

Proof. Exchanging the order of summation, $\bar{\beta} = \sum_{x\ne x'} p_x p_{x'} \Pr_k[h(k,x) = h(k,x')] \le (1-\mathrm{CP})/R$, valid for every p because the hypothesis is a statement about every pair $x \ne x'$; so $\gamma \le 1 - \mathrm{CP} - 1 = -\mathrm{CP}$. By (4) the conditional bound is at most $\frac{1}{2}\sqrt{R\cdot\mathrm{CP} - \mathrm{CP}} \le \frac{1}{2}\sqrt{R\,\epsilon_{h,z}}$, using $\mathrm{CP} \le \epsilon_{h,z}$. For the last claim, average over (H,Z) and apply Jensen's inequality to the concave nondecreasing $\sqrt{\cdot}$ — legitimate although p depends on H, since the conditional bound holds pointwise in (h,z) — together with $\mathbb{E}[\epsilon_{H,Z}] \le \epsilon$. □

The hypothesis is a property of a single table, whereas H in Section 1 is uniform on all of $\mathsf{Fun}(\mathcal{K}\times\mathcal{D},\mathcal{R})$; the second sentence of Proposition 2 is therefore about a different and more favourable model than the one at issue. It is stated to calibrate, not to apply. What it calibrates is this: on the first summand of (1) the leftover hash lemma gives $\frac{1}{2}\sqrt{\epsilon R}$, which is that summand with $c = \frac{1}{2}$. Nothing is lost there, and everything that separates the two routes therefore lies in γ.

4 The limitation

Proposition 2 does not apply here. Universality is not merely absent from a typical table; at the seed lengths at issue it is unavailable from any table, provided the output is shorter than the input.

Remark 1 (No universal family at small K, when $R < D$). If $\{h(k,\cdot)\}_{k\in\mathcal{K}}$ is 2-universal then, viewing each input x as the column vector $(h(k,x))_{k\in\mathcal{K}} \in \mathcal{R}^K$, two distinct columns agree in at most K/R coordinates, so the D columns form a code of length K over an alphabet of size R with minimum distance at least $(1-1/R)K$. The Plotkin bound, in the form "$d = (1-1/q)n$ implies $|C| \le qn$", gives $D \le RK$, that is $K \ge D/R$. So no 2-universal family exists once $K < D/R$, which at $D = 2^{256}$ and $R = 2^{64}$ holds for every $K < 2^{192}$.

The proviso matters. For $R \ge D$ a single injective row is already 2-universal, with $K = 1$, so no seed-length lower bound holds without a hypothesis relating R to D. An earlier version of this note asserted $K \ge D^{\Omega(1)}$ unconditionally, which that example refutes.

So γ must be taken over the supports the source may choose after reading H, and the theorem below says how large those make $R\bar{c}-1$. Recall that the source sees the entire table before *sd* exists, so any support it can name from H is admissible.

Theorem 1 (The collision route cannot beat $\sqrt{W/K}$). *Let $\mathcal{K},\mathcal{D},\mathcal{R}$ be nonempty and finite and let $\epsilon \in [1/D,1]$. Put*

$$t := \lceil 1/\epsilon \rceil, \qquad \sigma := \lceil Rt/D \rceil, \qquad W := \lfloor R/\sigma \rfloor.$$

Then $1 \le t \le D$, $1 \le \sigma \le R$ and $1 \le W \le R$, and for every *table $h \in \mathsf{Fun}(\mathcal{K}\times\mathcal{D},\mathcal{R})$ there is a source S — uniform on a set of t inputs, with $z=\perp$, and ϵ-unpredictable — for which*

$$R\,\bar{c}-1 \;\ge\; \frac{W-1}{K}, \qquad \text{so that the bound of Proposition 1 is at least} \qquad \frac{1}{2}\sqrt{\frac{W-1}{K}}.$$

No randomness is used: this holds for every table, not merely for most. When R and D/t are powers of two, as in the parameter ranges of interest, $W=\min\{R,\ D/t\}$ exactly; in general $W > \frac{1}{2}\min\{R,\ D/t\}-1$.

Proof. The parameters are in range. From $\epsilon \le 1$ we get $t \ge 1$, and from $\epsilon \ge 1/D$ that $t=\lceil 1/\epsilon\rceil \le D$; hence $Rt/D \le R$ and, R being an integer, $\sigma=\lceil Rt/D\rceil \le R$, while $\sigma \ge 1$ because $Rt/D>0$. From $\sigma \le R$ we get $W=\lfloor R/\sigma\rfloor \ge 1$, and from $\sigma \ge 1$ that $W \le R$.

The support. Fix any seed $k_1 \in \mathcal{K}$ and let $\mathcal{S}\subseteq\mathcal{R}$ consist of σ symbols r maximising $|\{x\in\mathcal{D} : h(k_1,x)=r\}|$. The D inputs are distributed among the R symbols, so the σ most popular of them carry at least a σ/R fraction of the total, that is at least $D\sigma/R$ inputs; and $\sigma \ge Rt/D$ gives $D\sigma/R \ge t$. So $\{x : h(k_1,x)\in\mathcal{S}\}$ has at least t elements; let T be any t of them and let $S(h)$ output $x \xleftarrow{\$} T$ together with $z \leftarrow \perp$. This is a counting fact about an arbitrary map $\mathcal{D}\to\mathcal{R}$; no property of h beyond being a function is used.

Admissibility. T is computed from h, which S receives in full, and *sd* is drawn only after S terminates, so nothing in the construction depends on the seed and S is a legitimate source for the game. Its conditional law is uniform on T, so $\epsilon_{h,\perp}=1/t=1/\lceil 1/\epsilon\rceil \le \epsilon$; since this holds for every table, $\mathbb{E}[\epsilon_{H,Z}] \le \epsilon$ and S is ϵ-unpredictable in the sense of Section 1.

The two collision bounds. For every k the law of $h(k,X)$ is a distribution on $\mathcal{R}$, so (2) with $A=\mathcal{R}$ gives $c_k \ge 1/R$. For $k=k_1$ every $x\in T$ has $h(k_1,x)\in\mathcal{S}$, so the law of $h(k_1,X)$ is carried by $\mathcal{S}$ and (2) with $A=\mathcal{S}$ gives $c_{k_1}\ge 1/\sigma$.

Assembling. Averaging the K lower bounds and using $R/\sigma \ge \lfloor R/\sigma\rfloor = W$,

$$R\,\bar{c}-1 \;\ge\; R\Big[\frac{1}{K}\cdot\frac{1}{\sigma}+\frac{K-1}{K}\cdot\frac{1}{R}\Big]-1 \;=\; \frac{R/\sigma}{K}+\frac{K-1}{K}-1 \;\ge\; \frac{W}{K}-\frac{1}{K} \;=\; \frac{W-1}{K},$$

and Proposition 1 applied to this source returns $\frac{1}{2}\sqrt{R\bar{c}-1} \ge \frac{1}{2}\sqrt{(W-1)/K}$.

The two forms of W. If $R=2^m$ and $D/t=2^\ell$ then $Rt/D=2^{m-\ell}$, so $\sigma=\max\{1,2^{m-\ell}\}$ and $W=\lfloor R/\sigma\rfloor=\min\{2^m,2^\ell\}=\min\{R,D/t\}$. In general $\sigma < Rt/D+1$, so, writing $a:=R$ and $b:=D/t$,

$$\frac{R}{\sigma} \;>\; \frac{a}{a/b+1} \;=\; \frac{ab}{a+b} \;=\; \frac{\min\{a,b\}\cdot\max\{a,b\}}{a+b} \;\ge\; \tfrac{1}{2}\min\{a,b\},$$

the last step because $a+b \le 2\max\{a,b\}$; hence $W > R/\sigma - 1 > \frac{1}{2}\min\{R, D/t\} - 1$. □

Remark 2 (Pinning more than one row). Confining j rows rather than one, each to σ symbols, gives $R\bar{c} - 1 \ge j(W-1)/K$ by the same averaging, provided a common support of size t survives. It does: choosing the j symbol sets $\mathcal{S}_1, \ldots, \mathcal{S}_j$ independently and uniformly among the subsets of $\mathcal{R}$ of size R/σ, each input survives all j constraints with probability σ^{-j}, so the expected number of survivors is $D\sigma^{-j}$ and some choice attains it; the constraint is $D\sigma^{-j} \ge t$, that is $j \le \log_\sigma(D/t)$. The gain is a factor $\sqrt{j}$ in the bound, hence $\log_2 j$ further bits of seed in the comparison below — one or two in the parameter ranges of interest — so the one-row construction of Theorem 1 is essentially the whole effect.

The averaging is the point. Choosing each $\mathcal{S}_i$ greedily, as the σ commonest symbols of its row, does *not* work for a worst-case table, since the j greedy sets may have a small or even empty common preimage; an earlier version of this note asserted the greedy version. Only the one-row case survives greedily, and that is the case Theorem 1 uses.

5 When the floor is a statement about γ

Theorem 1 bounds $R\bar{c} - 1 = R \cdot \mathrm{CP} + \gamma$ from below, and that is a bound on the sum. It carries no information about γ unless the floor exceeds what the first summand supplies on its own, which for the constructed source is exactly $R \cdot \mathrm{CP} = R/t$.

Lemma 1 (Attribution). *For the source of Theorem 1, $\mathrm{CP} = 1/t$ exactly, so $\gamma \ge (W-1)/K - R/t$. In particular, if*

$$\frac{W-1}{K} > \frac{R}{t}, \tag{5}$$

then the floor of Theorem 1 is not explained by the entropy deficiency, and $\gamma > 0$: the family is genuinely non-universal on that support, by a margin of at least $(W-1)/K - R/t$.

Proof. The source is uniform on T with $|T| = t$, so $\mathrm{CP} = \sum_{x \in T} t^{-2} = 1/t$. Substituting into (4) and rearranging Theorem 1's conclusion gives $\gamma = R\bar{c} - 1 - R/t \ge (W-1)/K - R/t$, and (5) makes it positive. □

Condition (5) is not automatic, and the comparison below is stated only where it holds. Where it fails — for instance $R = D$ and $\epsilon = 1/2$, so $t = 2$, $\sigma = 2$, $W = \lfloor R/2 \rfloor$, and $R/t = R/2$ exceeds $(W-1)/K$ for every $K \ge 1$ — the theorem is consistent with $\gamma \le 0$, that is with a universal family, and says nothing about the second summand.

Corollary 1 (The comparison, in seed length). *Fix a target error $\delta \in (0,1)$ and parameters for which (5) holds and the entropy deficiency is not itself binding, that is $c\sqrt{\epsilon R} < \delta$. Driving the second summand of (1) below δ requires $K \ge c^2 \log_2 D/\delta^2$, whereas by Theorem 1 the leftover-hash route cannot certify δ at all unless $K \ge (W-1)/(4\delta^2)$. In bits,*

$$\begin{aligned} \log_2 K &= 2\log_2 c + \log_2\log_2 D + 2\log_2(1/\delta) && \textit{(conjectured)}, \\ \log_2 K &\ge \log_2(W-1) - 2 + 2\log_2(1/\delta) && \textit{(collision route)}. \end{aligned}$$

Both carry $2\log_2(1/\delta)$; they differ in the first term, $\log_2 W$ against $\log_2\log_2 D$.

Side by side, and this is the whole comparison in one line:

$$\underbrace{c\left(\sqrt{\epsilon R} + \sqrt{\frac{\log_2 D}{K}}\right)}_{\text{conjectured, and proved in \texttt{proof.tex}}} \qquad \text{against} \qquad \underbrace{\frac{1}{2}\sqrt{\epsilon R + \frac{W-1}{K}}}_{\text{the least the collision route can return}} . \tag{6}$$

Since $\sqrt{a+b} \geq (\sqrt{a}+\sqrt{b})/\sqrt{2}$ by the arithmetic–geometric mean inequality, the right-hand side is at least $\frac{1}{2\sqrt{2}}(\sqrt{\epsilon R} + \sqrt{(W-1)/K})$, so the two are the same two ingredients up to constants and differ in exactly one place: what stands under the second square root. The conjecture pays $\log_2 D$, a number of bits; the collision route pays W, the size of a set.

Remark 3 (A worked instance). Take $D = 2^{256}$, $R = 2^{64}$, a source of min-entropy 160 (so $\epsilon = 2^{-160}$ and $t = 2^{160}$), $\delta = 2^{-32}$ and $c = 2$. Then $\sigma = \lceil 2^{-32} \rceil = 1$ and $W = R = 2^{64}$. Both hypotheses of Corollary 1 hold: the deficiency term is $c\sqrt{\epsilon R} = 2^{-47} \ll \delta$; and at the seed length (1) requires, $K = 2^{74}$, the floor is $(W-1)/K \approx 2^{-10}$ against $R/t = 2^{-96}$, so (5) holds with room to spare and Lemma 1 gives $\gamma > 0$. The comparison is then $\log_2 K = 74$ under (1) against $\log_2 K \geq 126$ for the collision route: a gap of 52 bits.

An earlier version of this note used min-entropy 128 here, where $c\sqrt{\epsilon R} = 2^{-31}$ exceeds $\delta = 2^{-32}$, so (1) could not certify that δ at any seed length and the comparison was empty. The hypotheses of Corollary 1 are not decorative.

Remark 4 (Where the loss comes from, and when it vanishes). The loss is the step (3), $\|v\|_1 \leq \sqrt{KR}\,\|v\|_2$, which is tight only when v is spread evenly over all KR cells; the source of Theorem 1 makes it far from that, its deviation on the pinned row being a single concentrated lump.

The true advantage against that source should not be mis-stated, and an earlier version of this note did mis-state it as $O(1/K)$. Since $\mathrm{SD}(Q,U) = \frac{1}{K}\sum_k \mathrm{SD}(\text{law}\, h(k,X), U_{\mathcal{R}})$, the pinned row k_1 contributes about $1/K$, but the other $K-1$ rows are *not* uniform. On a typical table each is t draws into R bins and contributes $\Theta(\sqrt{R/t})$, so the true advantage is $\Theta(\sqrt{R/t}) + \Theta(1/K)$, dominated by the *unpinned* rows in the regime of Remark 3; and on an adversarial table — every row constant, say — it is close to 1. So the overshoot of the lemma over the truth is neither $\sqrt{RK}$ nor uniform in the parameters. In the instance of Remark 3, at the seed length (1) requires, the lemma returns $\frac{1}{2}\sqrt{(W-1)/K} = 2^{-6}$ against a true advantage of about $0.4\sqrt{R/t} = 2^{-49.3}$; at the much larger seed length the lemma itself requires, the two come within a small factor of one another. The floor is a statement about what the technique can certify, not about how weak the source is.

The gap closes when W does. At $\epsilon = 1/D$ one has $t = D$, $\sigma = R$, $W = 1$ and Theorem 1 is trivial, correctly: a source of maximal entropy is uniform on all of $\mathcal{D}$ and has no support to select. More generally W is small whenever the source's min-entropy is close to $\log_2 D$ or the output is short: for $D = 2^{256}$, $R = 2^{64}$ and min-entropy 254 one has $t = 2^{254}$, $\sigma = 2^{62}$, $W = 4$, and the theorem exhibits no loss at all.

6 Reading the difference as key derivation

The game is a key-derivation problem. The input x is a secret with entropy but no uniformity — a Diffie–Hellman element, a biometric reading, the output of a physical source; sd is a public salt; H is a public hash; and $H(sd, x)$ is the derived key, which must look uniform to an adversary who knows H in full and sees sd. In that reading $\log_2 K$ is the *salt length* in bits, $m := \log_2 R$ the derived key length, $n := \log_2 D$ the length of the secret's encoding, and $\log_2(1/\epsilon)$ its min-entropy.

Whenever the derived key is no longer than the source's entropy gap, that is $m \le n - \log_2(1/\epsilon)$, one has $W = R$ and the two salt lengths of Corollary 1 read

$$\underbrace{\log_2 n + 2\log_2(1/\delta) + 2}_{\text{conjectured}} \qquad \text{against} \qquad \underbrace{m + 2\log_2(1/\delta) - 2}_{\text{collision route}}.$$

They differ in the first term, and they differ in kind. The conjectured salt grows with the *logarithm of the secret's length*; the collision route's salt grows with *the length of the key being derived.* So under the conjecture, deriving a 256-bit key rather than a 128-bit one from the same secret costs no extra salt whatever, while under the collision route it costs 128 further bits of salt. Concretely, with $c = 2$ and in each row the hypotheses of Corollary 1 verified:

	secret	min-ent.	key	error	salt needed (bits)		
	D	bits	bits	δ	conj.	route	univ.
DH element, 2048-bit group	2^{2048}	256	128	2^{-32}	77	190	1920
the same, 256-bit key	2^{2048}	512	256	2^{-64}	141	382	1792
a 512-bit source	2^{512}	300	128	2^{-64}	139	254	384

The last column is universal hashing, at $n - m$ bits from $K \ge D/R$ (Remark 1), and it is included because it makes the point of Section 7 concrete: universal hashing does not have a shorter salt than the conjecture, it has a far longer one. It is also read slightly differently from the other two. Those are the salt needed to reach the stated δ; this is the salt needed for such a family to *exist* at all, the binding constraint being existence rather than accuracy — at that salt the error is $\frac{1}{2}\sqrt{\epsilon R}$, which is 2^{-65}, 2^{-129} and 2^{-87} in the three rows, already inside δ in each.

The three columns scale in three different ways, and that is the whole comparison in one place:

$$\underbrace{\log_2 n + 2\log_2(1/\delta) + 2,}_{\text{conjectured}} \qquad \underbrace{m + 2\log_2(1/\delta) - 2,}_{\text{collision route}} \qquad \underbrace{n - m}_{\text{universal}}.$$

The conjectured salt grows with the logarithm of the secret's length; the collision route's grows with the length of the derived key; universal hashing's grows with the secret's length outright, *shrinks* as the derived key gets longer — 1920 bits for a 128-bit key against 1792 for a 256-bit one, since a larger R weakens the constraint $K \ge D/R$ — and does not depend on δ at all, because it is a structural requirement rather than an accuracy one. Only the first is what one wants of a salt: short, and growing only in the logarithm of anything.

Read the other way — at a fixed salt rather than a fixed error — the same gap says the route certifies almost nothing at the salt length the conjecture licenses. Substituting $K = c^2 \log_2 D/\delta^2$ into Theorem 1's floor gives

$$\frac{1}{2}\sqrt{\frac{W-1}{K}} \approx \frac{\delta}{2c}\sqrt{\frac{W}{\log_2 D}},$$

so the error the route can certify exceeds δ by a factor of about $\sqrt{W/\log_2 D}\,/\,(2c)$ — roughly $(m - \log_2 n)/2$ bits of security, over half the derived key's length. In the first row of the table that factor is $2^{56.5}$, so at a 77-bit salt the route certifies $2^{24.5}$: larger than 1, hence nothing at all. In the third row it certifies $2^{-6.5}$ at a 139-bit salt — not vacuous, but six bits of security where sixty-four were wanted.

Neither reading says that salting with 77 bits is unsafe. Both say that the collision route cannot be what establishes it, and Remark 4 is the reminder that the true advantage against the witness source is far below either bound. What is being measured here is the reach of a proof technique, not the security of a construction.

7 Is the random oracle worse than universal hashing?

Proposition 2 gives a universal family the bound $\frac{1}{2}\sqrt{\epsilon R}$ with no second summand at all, while Theorem 1 saddles a random table with a floor of $\frac{1}{2}\sqrt{(W-1)/K}$ on top of it. Read quickly, that says the random oracle is the worse extractor, which would be a strange thing for a random function to be. It is worth saying exactly why the reading is wrong, because the error is a natural one.

The two are not being compared at the same seed length. By Remark 1 a 2-universal family on D inputs needs $K \geq D/R$, a salt of $n - m$ bits. That is what buys $\gamma \leq 0$. The random-oracle question is posed at $K \approx c^2 \log_2 D/\delta^2$, a salt of $\log_2 n + 2\log_2(1/\delta)$ bits, which is smaller by an exponential factor. Comparing the two bounds without comparing their seeds is comparing the price of two objects while looking only at one price tag.

At equal seed length the random oracle matches, and the gap closes completely. Give the random table the salt a universal family requires, $K = D/R$. Then the conjectured second summand is

$$\sqrt{\frac{\log_2 D}{K}} = \sqrt{\frac{R \log_2 D}{D}},$$

which for $D = 2^{2048}$ and $R = 2^{128}$ is $2^{-954.5}$: not small, but absent. Both routes then return $\Theta(\sqrt{\epsilon R})$ and there is nothing to choose between them. The second summand is not a defect the random oracle carries and universal hashing escapes; it is the price of a *short* salt, and universal hashing never pays it only because it never has one.

At the seed length that makes the problem interesting, universal hashing does not exist. At the parameters of Section 6's first row — a Diffie–Hellman element in a 2048-bit group, min-entropy 256, a 128-bit key, $\delta = 2^{-32}$ — the conjectured bound does the job with a 77-bit salt. A universal family needs at least 1920 bits, twenty-five times as long, to do the same job no better: its error there is $\frac{1}{2}\sqrt{\epsilon R} = 2^{-65}$ against the random oracle's 2^{-63} at the same salt. Measured the way extractors are normally measured — error achieved per bit of seed — universal hashing is the seed-suboptimal construction and the random oracle is the optimal one. This is the standard picture, not a novelty of this setting: the leftover hash lemma is known to be far from optimal in seed length, and (1) sits at the information-theoretic optimum, which universal hashing misses by an exponential margin.

Why universality is immune to support selection is why it is expensive. This is the part worth keeping. The reason Proposition 2 survives a source that reads the whole table is visible in its proof: the bound $\bar{\beta} = \sum_{x \neq x'} p_x p_{x'} \Pr_k[h(k,x) = h(k,x')] \leq (1 - \mathrm{CP})/R$ holds for *every* law p, because universality constrains *every* pair $x \neq x'$ separately. A worst-case-over-all-pairs hypothesis is automatically robust to an adversary who picks the pairs last. But a worst-case-over-all-pairs hypothesis is exactly what Remark 1's counting argument charges D/R seeds for. The immunity and the cost are the same property seen twice. A random table satisfies only the average-case version, which is why choosing the support after seeing H can hurt it at all — and why $\log_2 n$ bits of salt suffice to stop that, rather than $n - m$.

And the floor is about the technique, not the object. Theorem 1 bounds what one particular accounting can certify about a random table. It is not a lower bound on the advantage, and the random table is in fact better than the floor suggests: (1) is *true*, proved in `proof.tex` with $c = 2$, so the random oracle does achieve $\sqrt{\epsilon R} + \sqrt{\log_2 D/K}$ at a short salt. What Theorem 1 says is that the collision route cannot be the proof of it. Remark 4 makes the same point numerically: against the witness source the true advantage is some 2^{-49} where the route returns 2^{-6}.

8 What is not claimed

Theorem 1 is one-sided. It says the collision route cannot return a bound better than $\frac{1}{2}\sqrt{(W-1)/K}$; it does *not* say the route attains that, and no matching upper bound is proved here. Every statement in this note about the route's cost is therefore a necessary condition on K, never the cost itself, and Corollary 1 is phrased that way deliberately.

The remaining window is wide and should not be papered over. Certifying γ for a random table by a bounded-differences argument — $\bar{\beta}$ has bounded-differences constant $2/(Kt)$ in the Kt table entries above a support of size t, and a union bound over the $\binom{D}{t}$ supports of that size costs $e^{t(1+\ln D)}$ — yields, for sources uniform on a support, $\gamma \leq R\sqrt{2(1+\ln D)/K}$ with high probability, hence an upper bound of $\frac{1}{2}\sqrt{\epsilon R} + \frac{1}{2}\sqrt{R}\,(2(1+\ln D)/K)^{1/4}$. Two gaps in that sketch are worth naming: it covers sources uniform on a support only, and a general source needs a further argument reducing to those; and the resulting ceiling exceeds Theorem 1's floor by a factor of order $\sqrt{R/W}\,(K \ln D)^{1/4}$, which in the instance of Remark 3 is about 2^{21}. So the truth for this route lies somewhere in that window, and locating it is not attempted.

What the floor alone does establish, in the regime where (5) holds and the deficiency term is not binding, is that the route cannot reach (1) there: it needs at least $\log_2(W-1) - 2\log_2 c - \log_2 \log_2 D - 2$ further bits of seed, which is 52 in the instance of Remark 3. Outside that regime nothing is claimed, and Remark 4 exhibits parameters — min-entropy 254 over the same D and R — where the construction shows no loss whatever.

Finally, the scope is one technique, not one lemma. Theorem 1 constrains the particular functional $\frac{1}{2}\sqrt{R\bar{c} - 1}$ of Proposition 1, arrived at by bounding the collision probability of $(sd, h(sd, X))$ and passing to statistical distance through (3). It says nothing about other uses of the leftover hash lemma, and the honest form of the headline is that (1) is not a corollary of *that route*. Nothing here is a statement about (1) itself, which is true: `proof.tex` proves it with $c = 2$, by bounding the L_1 distance directly and never forming a collision probability at all.